

===== Q1: HYBRID AI RECOMMENDATION ENGINE =====

Hybrid Recommendations:

I4 Score: 1.45

I2 Score: 1.433

I1 Score: 0.95

I3 Score: 0.85

Profile Agent: U1 behavior updated: Liked I2

Feedback Agent: U1 feedback = Positive

Top Recommendation: [('I4', 1.45), ('I2', 1.4333333333333333)]

===== Q2: SMART DISASTER DETECTION =====

Disaster Detection Results:

Chennai : HIGH FLOOD RISK

Chennai -> SEND EARLY WARNING

===== Q3: AI-BASED FRAUD DETECTION =====

Fraud Detection Results:

('T1', 0, 'SAFE')

('T2', 0, 'SAFE')

('T3', 0, 'SAFE')

('T4', 100, 'FRAUD')

('T5', 0, 'SAFE')

('T6', 100, 'FRAUD')

T1 -> NO ALERT

T2 -> NO ALERT

T3 -> NO ALERT

T4 -> FRAUD ALERT

T5 -> NO ALERT

T6 -> FRAUD ALERT

Learning Agent: Model updated using Confirmed fraud cases

===== Q4: SUPPLY CHAIN OPTIMIZATION =====

Total Sales:

P1 : 450

P2 : 200

P3 : 200

Inventory Decisions:

P1 : REORDER

P2 : REORDER

P3 : OPTIMAL

P1 Forecast: 495.0 Decision: REORDER

P2 Forecast: 220.0 Decision: REORDER

P3 Forecast: 220.0 Decision: STOCK OK

Logistics: DIRECT ROUTE

Procurement: SELECT LOW-COST SUPPLIER

===== Q5: PERSONALIZED LEARNING ANALYTICS =====

Student Average Scores:

S1 : 78.33

S2 : 68.33

S3 : 91.67

Weak Subjects:

S1 needs improvement in AI Score: 60

S2 needs improvement in Python Score: 55

Personalized Learning Recommendations:

Student: S1

Performance: 78.33

Status: Average

Recommended Content: Intermediate lessons + Quiz

Feedback: Good progress, continue practicing

Student: S2

Performance: 68.33

Status: Needs Improvement

Recommended Content: Basic lessons + Practice exercises

Feedback: Practice more and review weak topics

Student: S3

Performance: 91.67

Status: Good

Recommended Content: Advanced lessons + Projects

Feedback: Excellent performance

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CO5 DESIGN TASK COMPLETED

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Q1 - Hybrid Recommendation Engine

Content-Based + Collaborative Filtering + Agents

Q2 - Smart Disaster Detection

Multi-source Data Fusion + Disaster Agents

Q3 - Fraud Detection

Distributed Anomaly Detection + Fraud Agents

Q4 - Supply Chain Optimization

Demand + Inventory + Logistics + Procurement Agents

Q5 - Personalized Learning

Student Analytics + Adaptive Content + Feedback Agents

PySpark RDD Operations Used:

map()

filter()

flatMap()

groupByKey()

reduceByKey()

join()

persist()

The systems demonstrate:

- Distributed Processing
- Intelligent Agents
- Scalability
- Fault Tolerance
- Real-Time/Adaptive Decision Making